MGT 8040

GROUP 1

# *Optimizing Publications for Top Journals*

Presented by

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*Can measurable features of a scholarly paper reliably classify it as belonging to high impact or low impact journals?*

# *Updates from Last Revision*

## Classification

Initially, we made a linear model predicting numerical impfactor. Now, we classify impfactor into High and Low at the boundary of 7.

## AI → New Fields

Introduced a new categorical variable which investigates field differences. E.g. between only AI vs. Business vs. Business and AI.

## Sample Size

Our last iteration was done on 17 rows, and now we have expanded the search to 104 total observations spanning 3 years instead of 1.

*Three Main Stakeholders: PhD students, Faculty Members, and Educational Institutes*

# *Why are we doing this project?*

**+1.2 papers**

PhD students who publish during their doctoral studies show *higher research productivity by at least 1.2 papers.*

**85%**

85% of faculty members reach tenure as fast as 6 years with top rank publications.

**42%**

42% of PhD scientists and engineers work at educational institutions.


1. https://www.researchgate.net/publication/280301638_The_Impact_of_Publishing_During_PhD_Studies_on_Career_Research_Publication_Visibility_and_Collaborations
2. https://www.pnas.org/doi/10.1073/pnas.2500322122?
3. https://www.hbs.edu/ris/Publication%20Files/24-036_347e091f-838b-4ecc-b1a3-d793ea737188.pdf?

# Variables & Transformations

| Variable | Meaning / Description | Data Type |
|---|---|---|
| *Impact Factor* | Numeric rating of journal prestige | Continuous |
| *Page Count* | Total number of pages in the paper | Integer |
| *Word Count* | Estimated total word count | Integer |
| *Number of Authors* | Number of authors listed on the paper | Integer |
| *Number of figures* | Count of table and figures | Integer |
| *Paper Type* | 4 types:<br>review, experimental, conceptual, commentary | Categorical |
| *Number of references* | Total number of cited references in the paper | Integer |
| *Field* | 6 fields:<br>Business, Business & AI, Social Sciences, Social Science & AI, Healthcare, Healthcare & AI | Categorical |

## Added Variables

- Calculated using Formula: Citation Intensity & Visual Density
- Log-Transformed: References
- Interaction Terms: Field x Words

## Deleted Variables

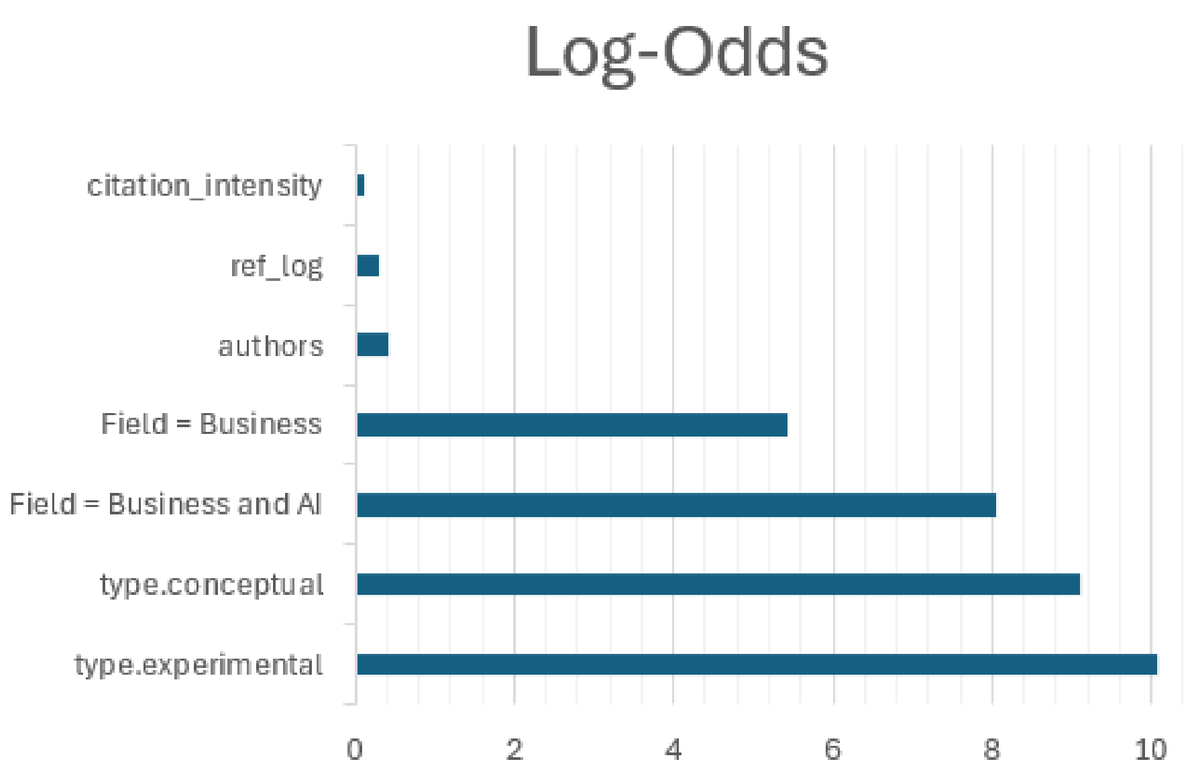
- Many Missing Values: Cite Score
- Multicollinearity with correlation > 0.7: Figures (with Visual Density) etc.
- Identity Variables: Title, URL, etc.

We applied 4 models to our dataset, prioritizing interpretation before anything. Logistic Regression comes out as the winner.

# *Modeling*

| Model | Accuracy | AUC | Specificity |
|---|---|---|---|
| Generalized Linear Model | 0.47 | 0.68 | 0.39 |
| Logistic Regression | 0.70 | 0.65 | 0.96 |
| Decision Tree | 0.60 | 0.50 | 0.86 |
| Random Forest | 0.60 | 0.61 | 0.71 |

| Model | Gains | Total Time |
|---|---|---|
| Generalized Linear Model | -41000 | 457 |
| Logistic Regression | -11000 | 377 |
| Decision Tree | -33000 | 457 |
| Random Forest | -15000 | 1248 |



# Confusion Matrix and Cost Benefit Analysis

**21**

True low Predicted low - model accurately predicted the papers which has low impact potential.
**Gain of $1,000 per case, $1000*21= $21,000**

**8**

False negative - certain high impact papers predicted as low impact papers.
**Loss of $7,000 per case, -($7,000)*8 = -($56,000)**

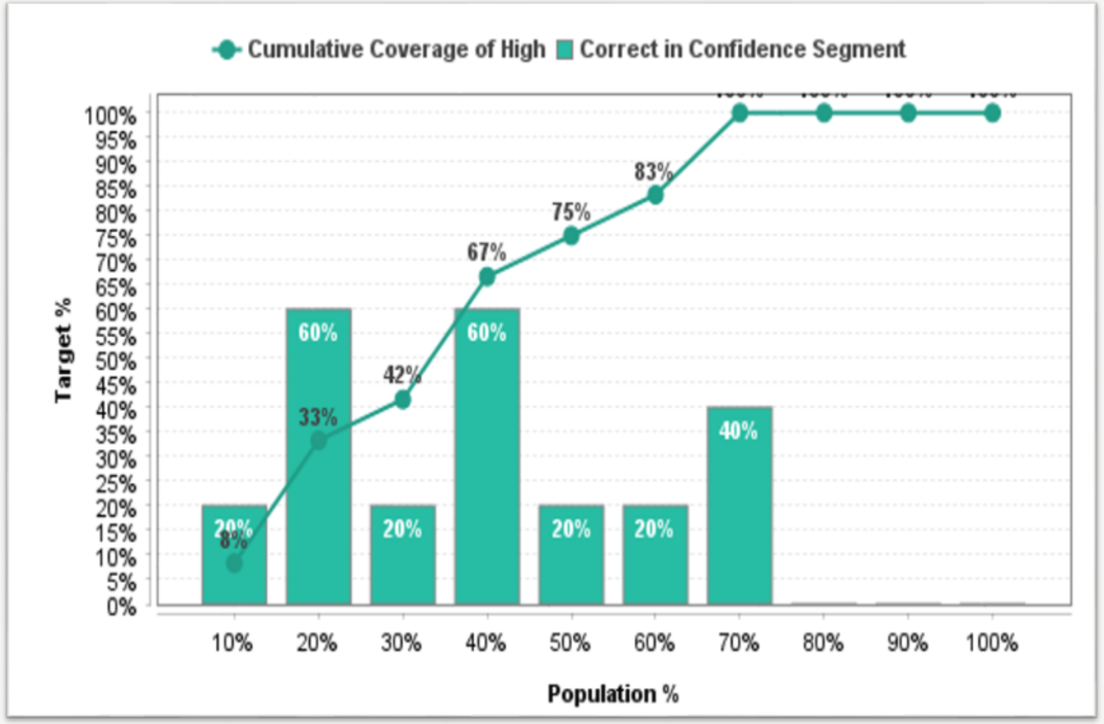
**1**

False positive - low impact paper predicted as high impact paper.
**Loss of $10,000 per case, -($10,000)*1= -($10,000)**

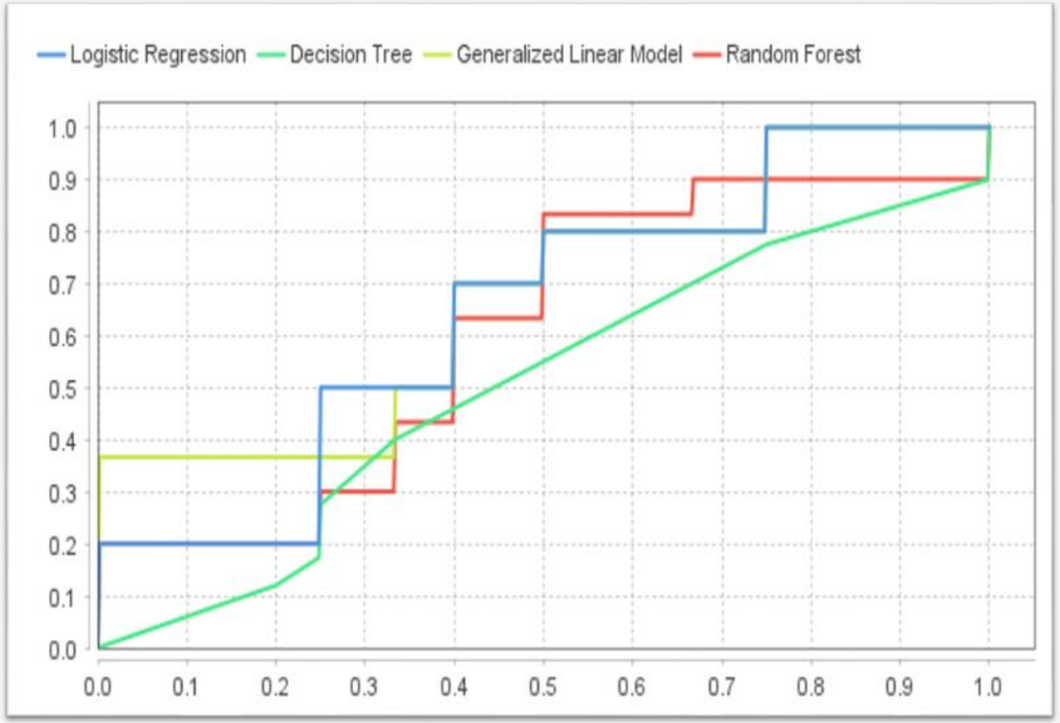
**0**

Model did not identify any paper as high impact paper. Weakness of model.
**Gain of $10,000 per case, $10,000*0= $0**

# ROC and Lift Chart




Lift Chart




ROC Comparison Curve

# Limitations & Future Work

## Limitations

- Small dataset
- Difficulty in identification (Sensitivity=0)
- Missing key features

- Collect more papers
- Funding data
- Include more features
- Tool to identify low impact papers

# *Thank you*

## For your attention